

Shruti S

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EDUCATION

VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI

Bachelor of Technology in Computer Science Engineering

Jun 2023-Present

CGPA 8.21

EXPERIENCE

VELLORE INSTITUTE OF TECHNOLOGY

Chennai, Tamilnadu

Intern - Research

JUN 2025-SEPT 2025

- Fine-tuned and evaluated a Large Language Model (LLM) to generate personalized, structured academic feedback, improving consistency and reducing manual evaluation effort.
- Engineered the end-to-end ML pipeline, including dataset curation, preprocessing, prompt optimization, evaluation, and iterative model refinement against faculty-defined quality metrics.
- Developed automated workflows for large-scale academic feedback generation, enabling scalable evaluation while maintaining output quality.
- Collaborated with faculty researchers to translate research objectives into deployable AI solutions through iterative experimentation and performance validation.
- **Stack:** Python · NLP · LLM Fine-tuning · Prompt Engineering · Dataset Curation · Applied AI

SHAKTA TECHNOLOGIES PVT LTD

Chennai, Tamilnadu

Intern - Structured Analysis & Risk Documentation

MAY 2025-JUN 2025

- Analyzed internal web applications and REST APIs to identify security, workflow, and process-level risks, producing structured technical reports with actionable recommendations.
- Collaborated with engineering and business stakeholders to translate technical findings into implementation-ready documentation for decision-making.
- Designed standardized assessment frameworks to evaluate application workflows, improving consistency in risk analysis across multiple systems.
- Documented functional requirements, system behavior, and engineering observations to support development and compliance initiatives.
- **Stack:** API Analysis · Technical Documentation · Risk Assessment · Requirement Analysis · Stakeholder Communication

PROJECTS

AGENTX – AUTONOMOUS BROWSER AI AGENT

[Link](#)

- Built an autonomous browser agent that converts natural language goals into executable task plans, enabling end-to-end web automation with planning, execution, verification, and self-correction.
- Engineered a modular agent architecture comprising an LLM-based planner, asynchronous execution engine, Playwright browser controller, verification pipeline, and SQLite-backed memory for reusable execution knowledge.
- Implemented adaptive failure recovery using retry, selector repair, and dynamic replanning strategies, allowing the agent to recover from DOM changes, navigation failures, and execution errors autonomously.
- Designed a benchmark suite spanning 25 browser automation tasks across navigation, extraction, form interaction, multi-step reasoning, and error recovery, achieving 88% overall task completion while averaging 15.9 s/task and 2.2 execution steps/task.
- **Stack:** Python · FastAPI · Playwright · SQLite · LLMs · AsyncIO · REST APIs

AUTHX – MULTI-LAYER CONTENT AUTHENTICATION & THEFT DETECTION SYSTEM

[Link](#)

- Developed a content authentication framework to detect image manipulation, AI-generated alterations, and partial-copy theft using perceptual hashing, frequency-domain analysis, and metadata verification.
- Achieved 97% detection accuracy on a curated evaluation dataset by combining pHash similarity scoring, DCT/FFT-based signal analysis, and adaptive thresholding while maintaining zero observed false positives.
- Designed a multi-stage verification pipeline integrating SHA-256 integrity checks, perceptual similarity matching, and forensic pattern analysis.
- Research accepted for presentation at the ICCSP 2026 peer-reviewed international conference.
- **Stack:** Python · SHA-256 · pHash · DCT/FFT Signal Processing · Pattern Analysis

PLOTLINES - EXPLAINABLE MOVIE RECOMMENDATION PLATFORM

[Link](#)

- Developed a full-stack movie recommendation platform leveraging Neo4j graph databases to model relationships between users, genres, actors, directors, and movies for personalized recommendations.
- Engineered an explainable recommendation engine using graph traversal algorithms and NLP-based preference extraction, generating recommendations aligned with user likes, dislikes, and viewing history.
- Designed RESTful APIs and optimized graph queries to efficiently retrieve connected movie recommendations while providing transparent reasoning for every suggestion.
- Built an interactive React-based interface enabling preference management, recommendation visualization, and real-time exploration of graph-based recommendations.
- **Stack:** TypeScript · React · Node.js · Neo4j · NLP · REST APIs

PUBLICATIONS

ICCSP'26

[Link](#)

A Hybrid AI-Blockchain Approach for Multi-Content Authentication and Ownership Proof Generation - Accepted, ICCSP 2026 (12th International Conference on Communication and Signal Processing 2026)

LEADERSHIP

IEEE COMPUTER SOCIETY

Chennai, Tamilnadu

Department Head

AUG 2024-Present

- Led a cross-functional student team of 50+ to plan and execute technical workshops, speaker sessions, and hackathons - improving department participation by 40%.
- Drove end-to-end strategic planning: from sourcing industry speakers and faculty advisors to managing logistics, timelines, and student engagement.
- Mentored junior members and built delegation systems that improved leadership continuity and team ownership.
- **Skills:** Team Leadership · Event Management · Stakeholder Coordination · Mentorship

SKILLS

- **Programming:** Python · JavaScript · Java · C · C++
- **Frameworks & Technologies:** FastAPI · Flask · React.js · Playwright · REST APIs · HTML5 · CSS3
- **AI & Machine Learning:** LLM Fine-tuning · Prompt Engineering · NLP · Computer Vision · Applied AI · Dataset Curation
- **Databases:** Neo4j · Supabase · SQLite · SQL
- **Developer Tools:** Git · GitHub · Postman · Figma
- **Core Concepts:** Data Structures & Algorithms · Object-Oriented Programming · DBMS · Operating Systems · Computer Networks · System Design · Asynchronous Programming

ACHIEVEMENTS & CERTIFICATES

- Research paper accepted at ICCSP 2026 (12th International Conference on Communication & Signal Processing) — on a Hybrid AI-Blockchain framework for content authentication.